MIL-R-22659D(AS)
15 February 1970
Superseding
MIL-R-22659C(AS)
15 February 1968

MILITARY SPECIFICATION

Radio Sets AN/ARC-51A, AN/ARC-51B, AN/ARC-51AX, and AN/ARC-51BX

This specification has been approved by the Naval Air Systems Command, Department of the Navy

1. SCOPE

1.1 Scope - The equipment covered by this specification shall provide two-way radio communication with properly equipped surface stations and other aircraft by means of amplitude-modulated radio telephony. When installed with the necessary associated equipment, the Radio Set shall be suitable for the following uses:

(1) Communication by means of A-M radio telephony (A3).

(2) Automatic direction finding.

(3) Automatic relaying wherein reception of signals on one equipment will cause transmission on another as outlined herein.

(4) Communication by means of Radio Telephone when use in conjunction with KY-28 equipment, KY-28 controls and the following receiver-transmitters:

Receiver-Transmitter, Radio RT-743B/ARC-51A
Receiver-Transmitter, Radio RT-780B/ARC-51AX
Receiver-Transmitter, Radio RT-767B/ARC-51B
Receiver-Transmitter, Radio RT-742C/ARC-51BX

1.2 Classification - The equipment covered by this specification shall consist of the following items:

FSC 5821

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Item	Type Designation	Appl. Para.
Receiver-Transmitter, Radio	RT-743/ARC-51A	3.4.1
Receiver-Transmitter, Radio	RT-780/ARC-51AX	3.4.1
Receiver-Transmitter, Radio	RT-767/ARC-51B	3.4.1
Receiver-Transmitter, Radio	RT-742/ARC-51BX	3.4.1
Receiver-Transmitter, Radio	RT-743B/ARC-51A	3.4.1
Receiver-Transmitter, Radio	RT-780B/ARC-51AX	3.4.1
Receiver-Transmitter, Radio	RT-767B/ARC-51B	3.4.1
Receiver-Transmitter, Radio	RT-742C/ARC-51BX	3.4.1
Control, Radio Set	C-6476/ARC-51A	3.4.2
Control, Radio Set	C-6555/ARC-51A	3.4.2
Control, Radio Set	C-6556/ARC-51B	3.4.2
Control, Radio Set	C-6287/ARC-51BX	3.4.2
Control, Radio Set	C-6300/ARC-51A	3.4.3
Control, Radio Set	C-6528/ARC-51B	3.4.3
Control, Radio Set	C-7307/ARC-51A	3.4.4
Control, Radio Set	C-8616/ARC	3.4.4
Mounting	MT-2653/ARC	3.4.5
Cooler, Air, Electronic Equipment	HD-513/ARC	3.4.6
	HD-615/ARC-51X	3.4.6
Indicator, Standing Wave Ratio	ID-1003/ARC	3.4.7
Indicator, Frequency Channel	ID-1472/ARC-51A	3.4.8
Indicator, Frequency Channel	ID-1752/ARC	3.4.8

1.3 Component Groups - Each Radio Set shall consist of one of the groups of components as listed below:

Radio Set	AN/ARC-51A	AN/ARC-51B	AN/ARC-51AX	AN/ARC-51BX
Radio Receiver-Transmitters	RT-743/ARC-51A or RT-743B/ARC-51A	RT-767/ARC-51B or RT-767B/ARC-51B	RT-780/ARC-51AX or RT-780B/ARC-51AX	RT-742/ARC-51BX or RT-742C/ARC-51BX
Radio Set	C-6476/ARC-51A or C-6555/ARC-51A or C-6300/ARC-51A or C-7307/ARC-51A	C-6556/ARC-51B or C-6528/ARC-51B or C-6287/ARC-51BX or C-8616/ARC	C-6476/ARC-51A or C-6555/ARC-51A or C-6300/ARC-51A or C-7307/ARC-51A	C-6287/ARC-51BX or C-6556/ARC-51B or C-6528/ARC-51B or C-8616/ARC
Mounting	MT-2653/ARC	MT-2653/ARC	MT-2653/ARC	MT-2653/ARC
Electronic Equipment Air Cooler	HD-513/ARC	HD-513/ARC	HD-615/ARC-51X	HD-615/ARC-51X
Standing Wave Ratio Indicator	ID-1003/ARC	ID-1003/ARC	ID-1003/ARC	ID-1003/ARC

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Frequency Channel Indicator	ID-1472/ARC-51A	ID-1572/ARC	ID-1472/ARC-51A	ID-1572/ARC

Radio Sets AN/ARC-51A and AN/ARC-51AX differ from Radio Sets AN/ARC-51B and AN/ARC-51BX respectively, in that the AN/ARC-51A and AN/ARC-51AX have a 600-ohm audio output impedance while the AN/ARC-51B and AN/ARC-51BX have a 150-ohm audio output impedance.

Radio Sets AN/ARC-51A, AN/ARC-51B and AN/ARC-51AX each have a 4 Kc audio bandwidth, while the AN/ARC-51BX has a 6Kc audio bandwidth. (See 3.4.2 and 3.4.3 for differences between Radio Set controls.)

1.4 Associated Equipment - This equipment shall operate with the following associated equipment which shall not be supplied as part of this equipment:

Antenna	AT-141A/AR or AT-450/ARC-45 with RG-8/U cable. Also UHF/L Band Blade Combination Antenna AS-1418/AR
Headset	H-1/AR or H-4/AR, H-173/AR
Microphone	T-17, M-92/U or ANB-M-C1 carbon type
Throttle Switch	NAF-1124-17
Intercommunication Set	AN/AIC-14, AN/AIC-12
Interconnecting Box	J-1013/AIC
Direction Finder Group	AN/ARA-25 or AN/ARA-50
KY-28 Control	C-8057/ARC
Security Equipment	KY-28

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MIL-C-172	Cases; Bases, Mounting; and Mounts, Vibration (For Use with Electronic Equipment in Aircraft)
MIL-W-5088	Wiring; Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment

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MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-C-6781	Control Panel; Aircraft Equipment, Rack or Console Mounted
MIL-P-7788	Plate, Plastic, Lighting
MIL-M-7793	Meter, Time Totalizing
MIL-C-8384	Connectors, Plug and Receptacle, Electrical (Molded Body) and Accessories, General Specification for
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303	Test Procedures, Preproduction and Inspection, For Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-C-26482	Connectors, Electric, Circular, Miniature, Quick Disconnect

STANDARDS

Military

MIL-STD-439	Electronic Circuits
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-781	Reliability Tests, Exponential Distribution
MS 28055	Aircraft Instrument Mounting Flange

PUBLICATIONS

ARINC Specification No. 410, "Mark 2 Standard Frequency Selection System"

EI-436	Avionics Installation Instructions for Radio Set AN/ARC-51A/B/AX/BX

2.2 Availability of Documents

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(1) When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Publications and Forms Center, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120

(2) Copies of ARINC Specification No. 410 may be obtained from ARINC, 1700 K Street, N.W., Washington, D. C.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

(1) Parts and materials requirements shall conform to Specification MIL-E-5400.

(2) Nonrepairable subassemblies, as outlined in Specification MIL-E-5400, shall be used when practicable. The general size of the subassembly, and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable. (See 6.5)

(3) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.1.1 Nonstandard Parts and Materials Approval - Approval for the use of nonstandard parts and materials including tubes, transistors, and diodes shall be obtained as outlined in Specification MIL-E-5400.

3.1.2 Switches - Rotary switches mechanically positioned by other than manual means shall be mounted by standoffs.

3.2 Design and Construction - The equipment shall conform with all the applicable requirements of Specification MIL-E-5400 for design, construction and workmanship, unless otherwise specified herein.

3.2.1 Operating Life

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3.2.1.1 Operational Stability - The radio set system shall operate with optimum performance for 300 hours, continuously or intermittently, without the necessity for readjustment of any controls which are unaccessible during flight.

3.2.1.2 Total Operating Life - The radio set system shall have a minimum total operating life of 5000 hours with reasonable servicing and replacement of parts. Parts requiring replacement within this interval and the life of such parts shall be specified by the manufacturer.

3.2.1.3 Time Totalizing Meter - A time totalizing meter conforming, as applicable, with Specification MIL-M-7793 shall be included in the RT-743/ARC-51A, RT-743B/ARC-51A, RT-767/ARC-51B, RT-767B/ARC-51B, RT-780/ARC-51AX, and RT-780B/ARC-51AX Receiver-Transmitter.

3.2.1.4 Reliability in Mean Time Between Failures (MTBF) - The radio set system shall have a minimum of 300 hours of mean (operating) time between failures when tested and accepted as outlined under the requirements of 4.4.3.

3.2.2 Cabling and Connection

3.2.2.1 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-E-5400.

3.2.2.2 Interconnection Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment.

3.2.3 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with the requirements of Specification MIL-N-18307.

3.2.4 Interference Control - The generation of radio interference by the radio set and the vulnerability of the radio set to radio interference shall be controlled within the limits of Specification MIL-I-6181 with the following exceptions:

(1) Antenna Conducted Keyup Condition - Spurious output between 200 and 500 mc shall not exceed 100 microvolts average and 400 microvolts maximum.

(2) Antenna Conducted Key Down Condition - The magnitude of all radiated signals from 200 to 500 mc, inclusive, on other than the carrier frequency

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shall be 40 db or more below the carrier signal. The magnitude of all radiated signals below 200 mc shall be 70 db or more below the carrier signal. The magnitude of all radiated signals above 500 mc shall be 80 db or more below the carrier signal.

(3) Receiver Intermodulation - The magnitude of the Receiver Intermodulation test signals shall be at least 80 db above one microvolt at the test sample terminal. Receiver sensitivity shall not be degraded by more than 2 db.

(4) Power Line Conducted Interference and Susceptibility - Frequency limits shall be 150 KC to 100 MC when testing the equipment. Receiver sensitivity degradation in this range shall be no greater than 2 db. For power line conducted interference, the limits at 25 mc shall apply over the 25 mc to 100 mc range.

(5) Radiated Interference - An increase of 16 db above the level stipulated in Specification MIL-I-6181 will be allowed in the frequency range from 80 to 1350 mc. The carrier output frequency may exceed this 16 db limitation by an additional 30 db.

(6) Susceptibility, Audio Frequency Conducted Receiver Requirements - The sensitivity of the receiver under standard conditions, which is 4 microvolts for transceivers with 4,000 Hz audio bandwidth and 4.2 microvolts for transceivers with 6,000 Hz bandwidth, shall be used as a reference level in performing the audio frequency conducted requirements of MIL-I-6181.

a. Power Line Conducted - Audio frequency conducted requirements of MIL-I-6181 shall apply to the 28 volt DC power line over the frequency range of 200 Hz to 15,000 Hz with the RT in the receive mode. Degradation of the main audio output signal-plus-noise-to-noise ratio shall not exceed 3 db.

b. Auxiliary Guard Conducted - Audio frequency conducted requirements of MIL-I-6181 shall apply to the auxiliary guard output terminals over the frequency range of 200 Hz to 15,000 Hz with the RT in the receive mode. Degradation of the main audio signal-plus-noise-to-noise-ratio shall not exceed 3 db.

Transmitter Requirements- The level of carrier modulation required to reach the clipping limit shall be used as a reference level in performing the audio frequency conducted requirements of MIL-I-6181.

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a. Power Line Conducted - Audio frequency conducted requirements of MIL-I-6181 shall apply to the 28 volt DC power line over the frequency range from 800 Hz to 15000 Hz with the RT in the transmit mode. Degradation of the modulated carrier signal-plus-noise-to-noise ratio shall not exceed 5 db.

b. Auxiliary Guard Conducted - Audio frequency conducted requirements of MIL-I-6181 shall apply to the auxiliary guard output terminals over the frequency range of 200 Hz to 15,000 Hz with the RT in the transmit mode. Degradation of the modulated carrier signal-plus-noise-to-noise ratio shall not exceed 3 db.

(6) Susceptibility, Audio Frequency Conducted - The normal sensitivity of the receiver, which is 4.0 microvolts for radio sets with 4000 cps bandwidth and 4.2 microvolts for radio sets with 6000 cps bandwidth, shall be used as a reference level in performance of this requirement.

(7) The limits for short duration interference caused by crystal over switching are relaxed by 20 db, with a maximum recurrence time of 4 cycles in three minutes permitted at room ambient temperature after a warm-up period of 15 minutes.

(8) The limits for short duration interference caused by channel changing operations are relaxed by 20 db, and the maximum duration of the interference is increased to 6 seconds.

3.2.5 Interchangeability - The equipment shall meet the interchangeability requirements as specified in Specification MIL-E-5400.

3.2.6 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in Specification MIL-E-5400. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance, except that only the antenna input and power connectors shall be mounted on the face of the equipment.

3.2.7 Control Panels - All rack or console mounted control panels shall conform to the applicable requirements of Specification MIL-C-6781. The configuration of all control panels must be approved by the procuring activity prior to preproduction testing.

3.2.8 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard

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conditions and for making laboratory bench tests.

Temperature	Room ambient (30°C ±10°C)
Altitude	Normal Ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input power voltage	115 ± 1.0 V AC and/or 27.5±0.5 V DC

3.2.9 Service Conditions - The equipment shall operate under any of the environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for Class I equipment except as modified herein.

3.2.9.1 Vibration - The equipment shall operate satisfactorily where subjected to the vibration requirements of Curves II and IV of Specification MIL-E-5400. Satisfactory performance when normally mounted (with vibration isolators in place) shall be demonstrated by compliance with the service condition requirements as specified herein. Satisfactory performance when mounted rigidly (without vibration isolators in place) shall be demonstrated by lack of mechanical failures and satisfactory operation under service condition limits after completion of the vibration test.

3.2.9.2 Altitude - The equipment shall operate satisfactorily at an altitude of at least 70,000 feet when supplied with external cooling air per Table 1 and 50,000 feet when supplied with ambient air from an attached blower. For the latter case, the ambient air temperature shall be as specified in Specification MIL-E-5400. Cooling air pressure drop shall be approximately as shown in Table 2.

Table 1

Cooling Air Requirements - Receiver-transmitter cooling air requirements at maximum system transmit-receive duty cycle in +71°C ambient air temperature when supplied from aircraft cooling system is:

Air Temperature (°C)	Flow Rate (lbs/hrs)
-15	27.2
- 5	27.5
5	29.0
15	32.3
25	37.8
35	46.4
45	58.0

Table 2

Cooling Air Pressure Drop - Cooling air pressure drop of each receiver-transmitter in nominal values is:

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Air flow lbs/hr	Pressure Drop, Inches of Water (Reference density 0.0765 lb/cu ft)
24	0.051
40	0.099
60	0.170
80	0.250
100	0.344

3.2.9.3 Acoustic Noise - No malfunction of the equipment shall occur when subjected to a jet engine exhaust noise of 120 db relative to 0.0002 dyne per square centimeter at frequencies between 20 and 10,000 cps.

3.2.10 Pressurization - Items of equipment required by this specification to be sealed shall comply with Specification MIL-E-5400 except that a desiccant is not required, a safety valve is not required, mechanical pressurization pumps shall not be used in the equipment and the cases shall be pumped with dry air to a differential of 2 to 4 pounds per square inch at sea level pressures and room ambient temperatures. Loss of pressure shall not exceed 5 pounds per square inch for a 24 hour period at 50,000 feet altitude and +55°C ambient temperature.

3.2.10.1 Air Pressure Indicator - A non-air release type pressure indicator shall be mounted in the front of the cover of the Receiver-Transmitter and shall give a visual indication when the internal air pressure is adequate for proper operation of the equipment.

3.2.11 Primary Input Power Requirements - The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance from the following power sources with characteristics as defined in MIL-STD-704 having limits as specified therein. The power required shall not exceed the specified amounts.

(1) RT-743/ARC-51A, RT-743B/ARC-51A, RT-767/ARC-51B and RT-767B/ARC-51B (including external blower)
   (a) AC Power (Three Phase) 115/200V, Category "B", 220 VA
   (b) DC Power 28V Category "B", 8 Amps
(2) RT-780/ARC-51AX and RT-742/ARC-51BX (including external blower)
   (a) DC Power 28V Category "B" 12.25 Amps
(3) RT-780B/ARC-51AX and RT-742C/ARC-51BX (including external blower)
   (a) DC Power 28V Category "B", 12.9 AMPS

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3.2.11.1 Overload Protection - A 15 ampere non-temperature sensitive fuse shall be provided in the primary power circuit of the AN/ARC-51AX and AN/ARC-51BX. An 8 ampere non-temperature sensitive fuse shall be provided in the DC primary circuit of the AN/ARC-51A and AN/ARC-51B.

3.2.11.2 Undervoltage Protection - The equipment shall not be damaged by voltages below the minimum specified in MIL-STD-704 when applied for a period of at least 15 minutes, and shall automatically resume normal operation when the voltage returns within limits.

3.2.12 Filament Protection - A protection device shall be incorporated in the equipment using zener diodes to maintain the filament voltage of the electron tubes at the proper operating levels when circuits may become unbalanced due to electron tube burnouts and module removals.

3.3 Performance

3.3.1 General - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply under standard conditions to both main and guard channel operation. Reduced performance and service requirements shall apply only where specifically outlined. Equipment performance under service conditions shall consist of a duty cycle wherein the equipment is operated on a continuous cycle of 10 minutes reception and 5 minutes transmission.

3.3.2 Warm-Up Time - The warm-up time for the equipment shall not exceed 5 minutes under standard conditions and 15 minutes at extreme service conditions.

3.3.3 Frequency Channels - By means of control box selection, it shall be possible to select any frequency in the range of 225.00 to 399.95 mc on 0.05 mc increments. These main channel frequencies shall correspond with the arithmetic series 225.00, 225.05, 225.10, 225.15----- 399.95 mc. Additionally, it shall be possible to select a fixed-frequency guard channel in the range of 238.00 mc to 248.00 mc. Unless otherwise specified, the equipment shall be supplied with the guard frequency set for 243.00 mc.

3.3.4 Terminations

3.3.4.1 Receiver RF - RF input to the receiver shall be through a 52-ohm cable form a signal generator having an effective source impedance of 52 ohms. Sensitivity shall be in terms of "open-circuit" microvolts.

3.3.4.2 Transmitter RF - The RF output load (dummy antenna) shall consist of a 52-ohm noninductive load which shall include means suitable for measuring the RF power output, percentage modulation and distortion.

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3.3.4.3 Receiver Audio - The main (headset) audio output channel of the receiver shall be terminated in either a 150-ohm or 600-ohm noninductive load which shall include means for measuring the audio-frequency power output. It shall be possible by changing transformer connections to match the main audio output impedance to either a 150-ohm or 600-ohm load, the 600-ohm termination being used in Radio Sets AN/ARC-51A and AN/ARC-51AX and the 150-ohm termination being used in Radio Sets AN/ARC-51B and AN/ARC-51BX. The auxiliary (wide range) audio output channel of the receiver shall be terminated in a resistance load of 20,000 ohms, including means for measuring the audio output voltage. The auxiliary guard output shall be terminated in a 600-ohm noninductive load, including means for measuring the audio output voltage.

3.4.4.4 Transmitter Input - The audio (microphone) input circuit of the transmitter shall be connected to an audio-frequency generator and dummy microphone having characteristics as listed below:

Carbon Microphone Input

Resistance - 82 ohms
Impedance (mainly resistive at audio frequencies) - 82 ohms
Frequency range - 150-20,000 cps
Total harmonic distortion - less than 1%
Open-circuit output potential (adjustable) 0.05 to 3.0 Vrms

Dynamic Mike Input

Resistance - 150 ohms
Impedance (mainly resistive at audio frequencies) - 150 ohms
Frequency range - 150-20,000 cps
Total harmonic distortion - less than 1%
The audio input circuit shall provide full performance capability when operating at a zero dbm level across 150 ohms. The audio input circuit to the input transformer shall be a balanced input type and shall be open circuited during the receive mode.

3.3.5 Sensitivity

3.3.5.1 4000- cps BW Sensitivity - The sensitivity of the receiver with a 4000-cps audio bandwidth shall be such that when an r-f input of 4 microvolts, 5 microvolts for guard receiver, (open circuit) modulated 30% at 1000 cps is applied, the audio output power shall be at least 25 milliwatts with a signal plus noise-to-noise radio (modulated to unmodulated) of 10 db or greater. When the main receiver is tuned to frequencies within +5 mc of the guard channel frequency, the main receiver and guard receiver shall meet the sensitivity requirement with 8 microvolts input. Under service conditions, the sensitivity of the main receiver

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shall not be worse than 7 microvolts, except that within +5 mc of the guard receiver the sensitivity shall not be worse than 10 microvolts. Under service conditions, the sensitivity of the guard receiver shall not be worse than 10 microvolts. Under special conditions of 22 volts dc and 102 volts ac power, the signal plus noise-to-noise ratio shall be at least 10 db for a 1000 microvolt signal modulated 30% at 1000 cps. With an RF input of 4 microvolts modulated 30% at 1000 cps, the signal plus noise-to-noise ratio of the auxiliary audio shall be 6 db or greater. With an RF input of 5 microvolts modulated 30% at 1000 cps, the signal plus noise-to-noise ratio of the auxiliary guard shall be 8 db or greater.

3.3.5.2 6000-cps BW Sensitivity - The sensitivity of the receiver with a 6000-cps audio bandwidth shall be such that when an r-f input of 4.2 microvolts, 5 microvolts for guard receiver, (open circuit) modulated 30% at 1000 cps is applied, the audio output power shall be at least 25 milliwatts with a signal plus noise-to-noise ratio (modulated to unmodulated) of 10 db or greater. When the main receiver is tuned to frequencies within +5 mc of the guard channel frequency, the main receiver and guard receiver shall meet the sensitivity requirement with 8.4 microvolts input. Under service conditions the sensitivity shall not be worse than 10 microvolts at any frequency. Under special service conditions of 22 volts dc and 102 volts ac power, the signal plus noise-to-noise ratio shall be at least 10 db for 1000 microvolt signal modulated 30% at 1000 cps.

With an RF input of 4 microvolts modulated 30% at 1000 cps, the signal plus noise-to-noise ratio of both the auxiliary audio and the auxiliary guard shall be 6 db or greater.

3.3.6 AVC Characteristics - For the RT-743/ARC-51A, RT-780/ARC-51AX, RT-767/ARC-51B, and RT-742/ARC-51BX receivers the AVC action shall be such that when a 1000 microvolt signal modulated 30% at 1000 cps is applied, the audio output power shall be 100 milliwatts (250 milliwatts for the RT-743B/ARC-51A, RT-780A/ARC-51AX, RT-767B/ARC-51B, and RT-742C/ARC-51BX) +20%, -10%. When the signal input (30% modulated at 1000 cps) is varied from 10 microvolts to 100,000 microvolts, the audio output power shall not vary by more than +3 db from the audio output obtained from the 1000-microvolt signal. When a main frequency within 5 mc of the guard frequency is selected, the foregoing requirement need be met only over the range 20 to 100,000 microvolts. Additionally, the receivers shall be protected against blocking under conditions of input signals up to one-half volt. (Blocking is defined as no audio power out). With 0.5 volt input, the audio output shall be not less than 25 milliwatts. The equipment shall be provided with an internal volume control which shall be adjusted so as to obtain the required output with the 1000-microvolt signal, and then locked before performing other receiver tests.

3.3.7 Selectivity - The over-all selectivity of the receivers shall fall within the limits set forth below for either standard or service conditions.

Total Bandwidth for RT-743/ARC-51A, RT-780/ARC-51AX, RT-767/ARC-51B and RT-742/ARC-51BX.

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Attenuation Below Resonant Response	Total Bandwidth Guard Receiver	Main Receiver
6 db	47 kc min.	47 kc min.
60 db	200 kc max.	85 kc max.

Total Bandwidth for RT-743B/ARC-51A, RT-780B/51AX RT-767B/ARC-51B and RT-742C/ARC-51BX main receivers.

Attenuation Below Resonant Response	Main Receivers
6 db	80 kc min.
60 db	150 kc max.

3.3.8 Images and Other Spurious Responses - When receiving on any channel, the receiver shall attenuate images and other spurious responses to the maximum practicable extent and in any case not less than 70 db relative to the desired signal, except that responses within 500 kc of the desired channel shall be attenuated not less than 45 db, and responses between 500 kc and 750 kc from the desired channel shall be attenuated not less than 55 db. For the guard receiver, spurious requirements for frequencies greater than 500 kc from the desired channel shall be governed by Specification MIL-I-6181.

3.3.9 I-F Rejection - Under either standard or service conditions, the receiver shall present an attenuation of not less than 80 db, relative to the response of the desired signal to signals on the frequencies of the intermediate frequency amplifiers.

3.3.10 Cross-Modulation and Desensitization - The cross-modulation and desensitization effects of interfering signals (modulated 30% at 1000 cps) on channels more than 500 kc from and at a level 60 db above the desired channel reference level (4 microvolts for the 4000 cps audio bandwidth radio sets and 4.2 microvolts for the 6000 cps audio bandwidth radio sets) shall permit a signal plus noise-to-noise ratio (desired signal) of at least 6 db. For interfering signals on channels within 500 kc of the desired channel the level shall be 30 db.

3.3.11 Regeneration - The interstage and over-all regeneration of the receiver shall be so minimized that no evidence of unnecessary oscillation, instability, or excessive regeneration shall occur at maximum r-f gain setting under either standard or service conditions.

3.3.11 Noise Peak Limiter - The detector of the receiver (following the intermediate frequency amplifier) shall be provided with an instantaneous noise peak limiter. The noise limiting action shall effectively limit the audio output voltage of modulating surges or peaks

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exceeding 50%+15% modulation of the r-f signal. Its action shall be rapid enough to prevent loss of intelligibility due to reduction in receiver gain under the influence of continuous r-f noises such as are caused by commutators, ignition systems, etc. When a pulsed signal having a pulse duration of 1 microsecond, a pulse repetition frequency of approximately 500 pulses per second, and peak amplitudes up to 0.1 volt is applied on the same frequency as a desired signal modulated 30% at approximately 1000 cps, not more than 50 microvolts input (desired signal) shall be required to obtain a signal plus noise-to-noise ratio of 6 db.

3.3.13 Audio Characteristics - The audio characteristics of the receiver shall be as set forth below. Except where otherwise stated, these requirements apply under both standard and service conditions, and apply to signals detected by either the fixed frequency or variable frequency receivers and amplified in the common headset circuit.

3.3.13.1 Capability -

(1) The RT-743/ARC-51A, RT-780/ARC-51AX, RT-767/ARC-51B and RT-742/ARC-51BX receivers shall be capable of the following:

Under standard conditions the receiver shall be capable of delivering an output of at least 100 milliwatts when a signal having any level between 10 and 10,000 microvolts modulated 30% at 1000 cps is applied. With a 1000 microvolt input signal modulated 30% at 1000 cps, the total harmonic content of the receiver output shall not be more than 10% at the 100 milliwatt level. Under service conditions the output shall not be more than 5 db below that obtained under standard conditions. Under the special service conditions wherein the supply voltage is 22 volts dc and 102 volts ac, an output of at least 14 milliwatts shall be obtained when a 10,000 microvolt signal modulated 30% at 1000 cps is applied.

(2) The RT-743B/ARC-51A, RT-780B/ARC-51AX, RT-767B/ARC-51B and RT-742C/ARC-51BX receivers shall be capable of the following:

Under standard conditions the receivers shall be capable of delivering an output of at least 250 milliwatts when a signal having any level between 10 and 10,000 microvolts modulated 30% at 1000 cps is applied. With a 1000 microvolt input signal modulated 30% at 1000 cps, the total harmonic content of the receiver output shall not exceed 10% at the 250 milliwatt level. Under service conditions the output

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shall not be more than 5 db below that obtained under standard conditions. Under the special service conditions wherein the supply voltage is 22 volts dc and 102 volts ac, an output of at least 35 milliwatts shall be obtained when a 10,000 microvolt signal modulated 30% at 1000 cps is applied.

3.3.13.2 Over-all Fidelity - The audio response shall not vary more than +1 or -3 db at 300 cps relative to the response at 1000 cps. It shall be possible by changing connections on a suitable terminal board to change the upper audio response relative to 1000 cps from +1 to -3db at 4000 cps to +1 to -3 db at 6000 cps. Modulation frequencies outside the 300 to 4000/6000 cps range shall be attenuated the maximum practicable extent without recourse to inclusion in the receiver or large and/or heavy filter components. Unless otherwise specified, the 300 to 4000 cps audio response shall be used in Radio Sets AN/ARC-51A, AN/ARC-51B, AN/ARC-51AX and the 300 to 6000 cps audio response shall be used in Radio Set AN/ARC-51BX.

3.3.13.3 Auxiliary Circuit - The auxiliary circuit shall deliver only signals detected by the variable frequency receiver. When a 1000-microvolt signal modulated 30% at 1000 cps is applied, the output shall be at least 0.25 volt, and when the modulation frequency is varied from 70 to 7000 cps, the output for any frequency in the range shall be within +3 db of the output obtained at 1000 cps. For the RT-743B/ARC-51A, RT-780B/ARC-51AX, RT-767B/ARC-51B and RT-742C/ARC-51BX the modulation frequency shall be varied from 70 to 20,000 cps and the output for any frequency in the range shall be within ±4 db of the output obtained at 1000 cps.

3.3.14 Volume Control - Operation of the volume control on the radio set control shall cause a minimum of change in the impedance which the radio set presents to external equipment attached to the headset terminals, and such impedance shall not be less that 50 ohms at any setting of the volume control. The volume control shall provide a minimum adjustment range of 26 db for the audio output from the receiver.

3.3.15 Carrier-to-Noise Ratio Squelch Characteristics - A carrier-to-noise ratio squelch circuit shall be incorporated in both the main channel and guard channel receivers. Internal squelch threshhold level controls shall be provided in both receivers. Under standard conditions the adjustment range of each control shall be such that the squelch threshold can be set to correspond to an output signal plus noise-to-noise ratio between the limits of 5 db and 10 db when the rf signal input of less than 4.0 microvolts is modulated 30 percent at 1000 cycles per second. The squelch circuit shall have positive action so that the receiver audio is either completely off or completely on at all times. Unless specified otherwise the squelch shall be adjusted to open at a 6 db signal pluse noise-to-noise ratio.

3.3.16 R-F Power Output - The transmitter shall be capable of delivering the following values of radio frequency carrier energy

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within the range described below:

Standard conditions	NLT 20 watts Average (16 watts min.)
Service conditions	7 watts min.
Reduced performance test at 102 vac and/or 22 vdc	3 watts min.

3.3.17 Modulated Characteristics - The transmitter shall meet the following performance requirements for modulation when operated under both standard and service conditions:

3.3.17.1 Radio Telephony (A-3)

3.3.17.1.1 Modulation - Under standard conditions the modulator shall be capable of modulating the r-f carrier output of the transmitter not less than 80% average when the input voltage at 1000 cps obtained from the dummy microphone, 3.3.4, is adjusted to 1 volt (open circuit) for the carbon microphone and to 0.8 volt (open circuit) for the dynamic microphone.

Modulation shall be determined from oscilloscope presentation of the modulated carrier wave and from measurement of maximum and minimum voltages displayed on the oscilloscope. Modulation shall be calculated from the formulas below:

Positive Peak Modulation	$\frac{E_{max} - E_o}{E_o} \times 100\%$
Negative Peak Modulation	$\frac{E_o - E_{min}}{E_o} \times 100\%$
Average Modulation	$\frac{E_{max} - E_{min}}{E_{max} + E_{min}} \times 100\%$

The modulator of this equipment shall employ clipper or limiter provisions for clipping positive and negative peaks at a modulation level of not more than 95% with increased microphone input voltage.

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The modulator gain control shall be adjusted to provide modulation to the clipping level with 1 volt (open circuit) input for the carbon microphone and 0.8 volt (open circuit) input for the dynamic microphone. With the input reduced 3 db below the clipping level, the distortion of 1000 cps modulation envelope shall not exceed 10%. At this level the transmitter shall have the following fidelity characteristics relative to the output at 1000 cps.

Relative output in db for the RT-743/ ARC-51A, RT-767/ARC-51B, RT-780/ARC-51AX and RT-742/ARC-51BX:

Frequency, cps	Relative Output in db
100	-5 or lower
300	+1 to -3
600	+1 to -3
1,000	0 (reference level)
3,000	+1 to -3
4,000 (RT-743/ARC-51A, RT-767/ARC-51B and RT-780/ARC-51AX)	+1 to -3
6,000 RT-742/ARC-51BX	+1 to -3

Relative output in db for the RT-743B/ ARC-51A, RT-767B/ARC-51B, RT-780B/ARC-51AX and RT-742C/ARC-51BX:

Frequency, cps	Relative Output in db
100	-5 db or lower
300	+4 to -4
600	+4 to -4
1,000	0 (reference level)
3,000	+2 to -2
4,000	+2 to -2
6,000	+3 to -3
10,000	+4 to -4
15,000	+4 to -4
20,000	+4 to -4

3.3.17.1.2 Modulation Under Service Conditions - Under service conditions the modulator shall be capable of modulating the r-f carrier output of the transmitter at least 70%. Distortion of the modulation envelope with the input 3 db below the clipping level shall not exceed 10%.

3.3.17.1.3 Microphone Current - The carbon microphone energizing current measured with dummy microphone connected and the equipment operated from a 27.5 volt d-c supply shall be 50± 7 milliampers.

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3.3.18 Frequency Stability - The frequency stability of the complete equipment after operation under either standard or service conditions after the required warm-up time shall be such as to maintain the equipment frequencies within the limits shown below provided the ambient temperature does not exceed 55°C:

Transmitter carrier frequency tolerance	±10 kc
Fixed - Frequency receiver resonant frequency tolerance	±13 kc
Variable - Frequency receiver resonant frequency tolerance	±10 kc

3.3.19 Sidetone - Monitoring of the carrier modulation shall be effected via the receiver audio output channel. The signals delivered to the receiver output while the transmission is in effect shall be gated by rectifying a small portion of the carrier output. Under standard conditions the minimum sidetone power output from the receiver shall be approximately 14 milliwatts when the transmitter carrier is modulated 30% at 1000 cycles. Under service conditions the output shall not be less than 10 milliwatts. The total harmonic content of the output shall not exceed 15%. A sidetone gate shall be provided which will open the sidetone circuit when the RF output falls to a level between 0.1 and 0.5 watts, to indicate to the operator the loss of adequate RF power output.

3.3.20 Internally Generated Noise - Under standard conditions and with no modulation voltage applied, the transmitter carrier noise shall be at least 40 db below the maximum modulation obtainable.

For the RT-743/ARC-51A, RT-780/ARC-51AX, RT-767/ARC-51B and RT-742/ARC-51BX receivers, the receiver signal plus noise-to-noise ratio shall be at least 30 db when a 1000-microvolt signal modulated 30% at 1000 cps is applied and the total receiver output (signal plus noise) is adjusted to be 100 milliwatts. Under service conditions, the receiver signal plus noise-to-noise ratio shall be at least 10 db when a 10-microvolt signal for the 4,000 cps audio bandwidth radio, or a 13 microvolt signal for the 6,000 cps audio bandwidth radio, modulated 30% at 1000 cps is applied.

For the RT-743B/ARC-51A, RT-780B/ARC-51AX, RT-767B/ARC-51B and RT-742C/ARC-51BX receivers, the receiver signal plus noise-to-noise ratio shall be at least 30 db when a 1000-microvolt signal modulated 30% at 1000 cps is applied and the total receiver output (signal plus noise) is adjusted to be 250 milliwatts.

The auxiliary audio and the auxiliary guard audio shall have a signal pluse noise-to-noise ratio of at least 25 db when a 1000 microvolt signal modulated 30% at 1000 cps is applied. The auxiliary audio shall have an output of at least 0.25 V and the auxiliary guard shall have an output in the range of 0.15 to 0.45 volts.

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3.3.21 Channel Changing Cycle - The time elapsing between the instant the channel selector switch is positioned to select a given channel and the instant the equipment is in an operative condition on the selected channel shall not be greater than 6 seconds under service conditions. This requirement shall hold for shifting the operating frequency of the equipment between any 2 channels. During channel changing operations, the output of the receiver (headset circuit) shall indicate that the equipment is not in readiness for transmission should the operator begin speaking before the channel changing cycle is completed. During channel change, the audio output of the receiver-transmitter shall be capable of being reduced to zero, or an audio tone shall be applied to the output. This tone shall be 400 cps in Radio Set AN/ARC-51A and AN/ARC-51B and 800 cps in Radio Sets AN/ARC-51AX and AN/ARC-51BX. It shall be possible by changing connections on a suitable terminal board to select either means of indicating the channel changing cycle. Unless otherwise specified, the tone shall be applied during the channel changing cycle.

3.3.22 Frequency Modulation - Under the service conditions and with no modulating voltage applied to the transmitter and no r-f signal applied to the receiver the peak frequency deviation arising from frequency modulation of the receiver conversion oscillator or the transmitter carrier shall not exceed 1.0 kc. Under the standard conditions, and with the emitted carrier modulated 80% at 1000 cycles, the peak frequency deviation arising from frequency modulation of the transmitter shall not exceed 1.0 kc.

3.3.23 Transmit-Receive Interval - When shifting from transmission to reception or vice versa on the same channel (with an r-f signal input voltage of 1000 microvolts modulated 30%), not more than 100 milliseconds shall elapse between the operation of the control switch and the time the equipment is stabilized in the selected mode of operation in compliance with the performance requirements. This requirement shall be met under both standard and service conditions.

3.3.24 Accuracy of Automatic Tuning - When the same frequency channel is selected several times in succession, the transmitter r-f power output and the receiver audio power (with an input signal of 100 microvolts or greater modulated 30% at 1000 cycles) shall not vary by more than 0.5 db from the power outputs obtained initially. This requirement shall be met under standard or service conditions provided such conditions are static.

3.3.25 Cable Length - Under either standard or service conditions, the equipment shall operate as required by this specification when using control cables up to 25 feet in length fabricated from #22 AWG copper wire, or when using cables up to 110 feet in length fabricated from #16 AWG copper wire.

3.3.26 Relaying - When two equipments are connected for automatic relaying, the performance shall be as follows:

3.3.26.1 Single-Channel Communication - When transmission and reception only are selected on a single set (either receiver-transmitter),

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performance shall be as required by this specification for one receiver-transmitter in a nonrelaying installation.

3.3.26.2 Dual Reception, Single Transmission - When reception on both sets and transmission on one set is selected, the performance rendered by the sets shall be as required by this specification for one set operating singly in the mode of operation selected, (transmission or reception) except that the receiver audio power output (or sidetone) may decrease not more than 2 db.

3.3.26.3 Automatic Relaying Operation

(1) Voice Source Information - The application of an RF signal to the antenna circuit of either radio transceiver, on the receiver resonant frequency (F_1), of a sufficient level to fully open the receiver squelch circuit (6 db signal pulse noise-to-noise ratio) shall key the transmitter of the other transceiver and generate an amplitude modulated RF carrier of the rated power on a selected frequency (F_2), and removal of the actuating RF signal shall unkey the transmitter and return the transceiver to receive condition.

When the actuating signal is 3 db or more above the threshold signal plus noise-to-noise ratio where relaying is possible, then the transmitter shall reach its rated power output within 150 milliseconds after application of the actuating RF signal, and the transceiver shall be returned to receive condition within 150 milliseconds after removal of the actuating signal.

(2) Cypher Source Information - The application of an actuating RF signal to the antenna circuit of either radio transceiver, on the receiver resonant frequency (F_1), of sufficient level to generate a detected auxiliary audio output of 8 db signal plus noise-to-noise ratio (threshold), shall key the transmitter of the other transceiver and generate an amplitude modulated RF carrier at the rated transmitter power output on the selected frequency (F_2). Within 150 milliseconds after application of the actuating RF signal, the transmitter shall reach full rated output, and removal of the actuating signal shall return the transceiver to receive condition within 150 milliseconds.

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3.3.26.4 Automatic Relaying Modulation

(1) Voice Source Information - The transmitter of either transceiver shall be amplitude modulated a minimum of 80% by utilizing the main audio output from the other transceiver's receiver, appropriately voltage divided and impedance transformed, as defined in paragraph 3.4.3.5.1 contained herein. For test purposes, when an RF signal amplitude modulated 80% at 1000 Hz (3 db above the level required to actuate the squelch adjusted to open at 6 db signal plus noise-to-noise ratio) is applied, the transmitted carrier shall be amplitude modulated a minimum of 80%.

(2) Cypher Source Information - The transmitter of either transceiver shall be amplitude modulated a minimum of 80% by utilizing the auxiliary audio output from the other receiver, appropriately amplified and impedance transformed, as defined in paragraph 3.4.3.5.1.

For test purposes, when an RF signal amplitude modulated 80% at 1000 Hz (3 db above the level required to key the transmitter, with the auxiliary audio threshold adjusted to 8 db signal plus noise-to-noise ratio) is applied, the transmitted carrier shall be amplitude modulated a minimum of 80%.

3.3.26.5 Relaying Monitor - When the equipment is relaying as described above, the operator headsets of both equipments shall be connected so as to monitor all signals relayed in either direction. The operator of either equipment can disable the automatic relaying operation and transmit by pressing the push-to-talk switch. Under these conditions, the audio power output may decrease not more than 6 db. Distortion and signal plus noise-to-noise ratio at the terminals of the headset circuit, and r-f carrier output and modulation shall be essentially the same as those obtained when ordinary operation is selected. Incorporate in the C-6300/ARC-51A provisions to amplify an AUX GUARD input (.2 VRMS minimum across 600 ohms) to a 12 VRMS output across 600 ohms. Provide capability in cypher relay mode to utilize the C-6300/ARC-51A volume control to attenuate either an externally applied KY-28 "audio in" (12 VRMS across 600 ohms) or an internally applied AUX GUARD "amplified audio" 12 VRMS across 600 ohms.

3.3.27 Automatic Direction Finding - When installed with certain associated equipment, the radio-set shall be suitable for automatic direction finding. When used for this service, the receiver-transmitter will receive radio frequency signals from the direction finding antenna, amplify and demodulate them and deliver the low-frequency modulation components to the direction finder amplifying and indicating equipment. When the radio set is connected in an automatic relaying installation, the

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automatic direction finding function is available on one receiver-transmitter. When the relaying function is selected on the C-6300/ARC-51A or C-6528/ARC-51B, the ADF function is disabled. The auxiliary audio output (see 3.3.13.3) shall be in phase +0.28 milliseconds with the negative portion of the rf envelope of a 1000 microvolt signal modulated with a 200 cps square wave.

3.3.28 Frequency Selection System - The equipment shall be designed to use the 2 out of 5 code system in accordance with ARINC Specification No. 410 "Mark 2 Standard Frequency Selection System" dated 1 October 1961.

3.4 Detailed Requirements

3.4.1 Receiver-Transmitter Radio RT-743/ARC-51A, RT-767/ARC-51B, RT-780/ARC-51AX, RT-742/ARC-51BX, RT-743B/ARC-51A, RT-767B/ARC-51B, RT-780B/ARC-51AX and RT-742C/ARC-51BX - The Receiver-Transmitters shall meet the following requirements:

3.4.1.1 Function - The receiver portion shall receive r-f energy from the antenna, amplify and convert it to aural intelligence. The transmitter portion shall generate r-f power and transmit it to the load specified herein. The receiver-transmitter shall be capable of operation with their respective control boxes as listed in 1.3.

3.4.1.2 Form Factor and Mounting - The receiver-transmitter shall be contained in a single sealed case meeeting the applicable requirements of Specification MIL-C-172 and capable of being mounted in an aircraft by means of Mount MT-2653/ARC. It shall be possible to dismount and remove it from its case using ordinary hand tools and without requiring removal of any cables other than the coaxial and power connectors. Vibration isolators shall be arranged to furnish all support required by the receiver-transmitter and shall provide space between the mounting rails for the external air ducts which are part of the sealed case assembly. The over-all dimension of the receiver-transmitter shall not exceed 7 inches in height, 9-9/16 inches in width and 15-1/8 inches in length. All modules and corresponding chassis space locations shall be clearly identified by part number and name.

3.4.1.2.1 Warning Notice and Mechanical Clearance - The following notice shall be placed on the plenum assembly: "Mounting surface of modules must be flush with chassis before tightening mounting screws. Adequate clearance shall be provided to allow for ease of adjustment of all circuits during electrical alignment procedures."

3.4.1.3 Contents of the Unit - This unit shall contain the complete transmitter, modulator, main receiver, guard receiver, direct crystal-controlled frequency generator, frequency selection system, automatic tuning devices, and all other parts to make the equipment function as a unit except the radio set control specified herein. It shall be possible to remove or replace any plug-in unit located on the main chassis using ordinary hand tools and without unsoldering any connections. It

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shall be further possible to replace any plug-in unit with another without resynchronization except that realignment of tuning elements may be necessary. The RT-743/ARC-51A, RT-743B/ARC-51A, RT-767/ARC-51B and RT-767B/ARC-51B shall contain power conversion facilities which permit operation from the primary power source described in 3.2.11(1). The RT-780/ARC-51AX and RT-742/ARC-51BX shall contain power conversion facilities which permit operation from the primary power source described in 3.2.11(2). The RT-780B/ARC-51AX and RT-742C/ARC-51BX shall contain power conversion facilities which permit operation from the primary power source described in 3.2.11(3). An internal blower shall be used to provide circulation of cooling air within the pressurized portion of the equipment. The internal blower for the AN/ARC-51A and AN/ARC-51B shall operate from the 115 V, 400 cps power supply. The internal blower for AN/ARC-51AX and AN/ARC-51BX shall operate from the 55 V, 800 cps single phase power supply contained within the equipment. Power for the external blower shall be furnished through the external connector.

3.4.1.4 Weight - The weight of the receiver-transmitter shall not exceed 29.1 pounds for RT-743/ARC-51A, RT-767/ARC-51B, RT-743B/ARC-51A and RT-767B/ARC-51B and 29.2 pounds for RT-780/ARC-51AX, RT-742/ARC-51BX, RT-780B/ARC-51AX and RT-742C/ARC-51BX.

3.4.1.5 Electrical Connections

3.4.1.5.1 External - The following electrical connections shall be provided on the receiver-transmitter:

(1) Coaxial connector, Type N, for connection of receiver-transmitter to the antenna circuits.

(2) Two connectors conforming to Specification MIL-C-26482 shall be provided on the front of the equipment for connection of power and control circuits.

(3) A coaxial connector, Type N, shall be used on the output of the reflectometer filter.

3.4.1.5.2 Internal - Module power and signal connectors, shall be of a quality conforming to the specification requirements of Specification MIL-C-8384 except that contact retention shall be 1 oz. minimum and 8 oz. maximum. Coaxial connectors shall meet the performance requirements of the equipment specification and may be an integral part of the module power and signal connectors.

3.4.1.6 Controls, Internal - The following semi-fixed adjustable controls shall be accessible with the unit removed from its case:

(1) Main receiver sensitivity control
(2) Guard receiver sensitivity control
(3) Main audio amplifier gain control

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(4) Guard audio amplifier gain control

(5) Modulator audio gain control

(6) Squelch control

3.4.2 Controls, Radio Set C-6476/ARC-51A, C-6555/ARC-51A, C-6556/ARC-51B, and C-6287/ARC-51BX - These controls shall conform to the applicable requirements of Specification MIL-C-6781 for Type I controls and shall be suitable for installation in a console panel.

3.4.2.1 Function - Radio Set Controls C-6476/ARC-51A, C-6555/ARC-51A, C-6556/ARC-51B and C-6287/ARC-51BX shall provide prime control of the various radio set configurations as indicated in 1.3. Radio Set Control C-6476/ARC-51A shall match the 600-ohm audio output impedance of Receiver-Transmitters RT-743/ARC-51A, RT-780/ARC-51AX, RT-743B/ARC-51A, and RT-780B/ARC-51AX, and requires 28 volts for panel illumination. The C-6555/ARC-51A shall match the 600-ohm audio output impedance of RT-743/ARC-51A and RT-743B/ARC-51A and requires 5 volts for panel illumination. The C-6556/ARC-51B shall match the 150-ohm audio output impedance of RT-767/ARC-51B and RT-767B/ARC-51B and requires 5 volts for panel illumination. The C-6287/ARC-51BX shall match the 150-ohm audio output impedance of RT-742/ARC-51BX and RT-742C/ARC-51BX and requires 28 volts for panel illumination. It shall be possible to manually select any of the available frequencies and preset up to 20 channels on any of these available frequencies. The unit shall have red lighting within the plastic edge-lighted plate in accordance with Specification MIL-P-7788.

3.4.2.2 Form Factor and Mounting - The over-all dimensions of this unit shall not exceed 5-3/4 inches in width, 4-7/8 inches in height and 6-1/4 inches in depth.

3.4.2.3 Weight - The weight of this unit including the mating connector shall not exceed 3.8 pounds.

3.4.2.4 Contents of the Unit - This control shall contain all circuits necessary to control manual frequency or preset channel selection, emission, ADF operation and primary power to the equipment.

3.4.2.5 Electrical Connections - All electrical connections to this unit shall be made through a connector conforming to Specification MIL-C-26482.

3.4.2.6 Controls, Panel-Mounted - The following controls shall be located and suitably marked on the front panel.

(1) A 20-position rotary switch-type control "CHAN" for selecting any one of 20 predetermined main frequencies.

(2) A 4-position rotary switch for controlling all primary input power to the equipment and for selecting the mode of transmission and reception as follows; Position One - "OFF,"

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Position Two - "T/R" operation on frequency selected; Position Three - "T/R+G" - guard receiver operation in addition to that of Position Two; Position Four - "ADF" operation on frequency selected.

(3) A rotary volume control "VOL" for adjusting the amplitude of the audio signals delivered to the headset circuit.

(4) Three controls operating a digital display "Megacycles" for manually selecting any one of 3500 channels. The frequency readout display shall indicate the frequency on which the equipment is operating when in the "MAN" position. Provisions shall be included on the C-6287/ARC-51BX to block out the hundredths megacycle digit for operation with 1750 channel equipment.

(5) A 3-position switch to allow rapid selection of three operating frequencies as follows: Position One - "PRESET", for operation on any one of the predetermined frequencies selected by (1) above; Position Two - "MAN", for operation on any of the manually selected frequencies determined by (4) above; Position Three - "GUARD XMIT", for operation on the guard frequency.

(6) Means for readily setting up new predetermined frequencies on the 20-position switch type control.

(7) A "SQ DISABLE" switch for disabling the squelch.

3.4.2.7 Vibration - Maximum vibration acceleration to which this control unit shall be subjected shall be governed by curve IV of Specification MIL-E-5400.

3.4.3 Controls, Radio Set C-6300/ARC-51A and C-6528/ARC-51B - These controls shall conform to the applicable requirements of Specification MIL-C-6781 for Type I controls and shall be suitable for installation in a console panel.

3.4.3.1 Function - Radio Set Control C-6300/ARC-51A shall provide prime control of Radio Set AN/ARC-51A or AN/ARC-51AX and shall include all controls for operation and frequency selection of Receiver-Transmitter RT-743/ARC-51A, RT-780/ARC-51AX, RT-743B/ARC-51A and RT-780B/ARC-51AX when used with the C-6476/ARC-51A in an automatic relaying installation. The C-6300/ARC-51A shall match the 600-ohm audio output impedance of RT-743/ARC-51A, RT-780/ARC-51AX, RT-743B/ARC-51A and RT-780B/ARC-51AX. Radio Set control C-6528/ARC-51B shall provide prime

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control of Radio Set AN/ARC-51B and shall include all controls for operation and frequency selection of Receiver-Transmitter RT-767/ARC-51B and RT-767B/ARC-51B when used with the C-6556/ARC-51B in an automatic relaying installation. The C-6528/ARC-51B shall match the 150-ohm audio output impedance of RT-767/ARC-51B and RT-767B/ARC-51B and requires 5 volts for panel illumination. During automatic relaying operation the headset audio is common for #1 and #2 equipments. Volume controls on the control units adjust the individual headset audio levels. The operator of either equipment can disable the automatic relaying operation and transmit by pressing the push-to-talk switch. It shall be possible to manually select any of the available frequencies and preset up to 20 channels on any of these available frequencies. The unit shall have red lighting within the plastic edge-lighted plate in accordance with Specification MIL-P-7788. Either 28 VDC or 5 VDC red panel lighting shall be provided in accordance with MIL-P-7788. Lamps shall be replaced from the front panel. For shipment install 28 VDC lamps for shipment and provide 5 VDC lamps as spares.

3.4.3.2 Form Factor - The over-all dimensions of the unit shall not exceed 4-7/8 inches in length, 5-3/4 inches in width and 7-1/4 inches in depth.

3.4.3.3 Weight - The weight of this unit including the mating connector shall not exceed 4.1 pounds.

3.4.3.4 Contents of the Unit - This control unit shall contain all circuits necessary to control manual frequency or preset channel selection, emission, ADF operation, automatic relaying operation and primary power to the equipment.

3.4.3.5 Electrical Connections - All electrical connections to this unit shall be made through a connector conforming to Specification MIL-C-26482.

3.4.3.5.1 Terminations

(1) Voice Source Information - The C-6300/ARC-51A when used to relay voice source information shall accept the main receiver audio output (maximum of 12.1 volts across 600 ohms impedance) and convert this audio output to a nominal level of 1 volt rms (open circuit) across a 100 ohm impedance. The level and impedance conversion shall not degrade the fidelity of the original signal.

(2) Cypher Source Information - The C-6300/ARC-51 when used to relay cypher source information shall accept the receiver auxiliary audio output (.25 Vrms minimum across 20,000 ohms impedance) and convert this audio output to a nominal level of 1 volt rms (open circuit) across a 100 ohm impedance. The level and impedance conversion

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shall not degrade the fidelity of the original signal.

3.4.3.5.2 Voice/Cypher Relay Mode Switch - The C-6300/ARC-51A shall have provisions, while in the Relay Mode, for operation with a Voice/Cypher Relay Mode Switch (contained on C-6300/ARC-51A control) whereby the operator may manually switch from voice relay to cypher relay. In the Cypher Mode, provisions to ground the receiver main audio in the C-6300/ARC-51A shall be incorporated.

3.4.3.6 Controls, Panel-Mounted - The following controls shall be located and suitably marked on the front panel.

(1) A 20-position rotary switch-type control "CHAN" for selecting any one of 20 predetermined main frequencies.

(2) A 5-position rotary switch for controlling all primary input power to the equipment and for selecting the mode of transmission and reception as follows: Position One - "OFF"; Position Two- "T/R" operation on frequency selected; Position Three - "T/R+G" guard receiver operation in addition to that of Position Two; Position Four - "ADF" operation on frequency selected; Position Five - "REL/P" operation on frequency selected; Position Six - "REL/C" operation on frequency selected.

(3) A rotary volume control, "VOL", for adjusting the amplitude of audio signals delivered to the headset circuit.

(4) Three controls operating a digital display "MEGACYCLES" for manually selecting any one of 3500 channels. The frequency readout display shall indicate the frequency on which the equipment is operating, when in the "MAN" position.

(5) A three-position switch to allow rapid selection of three operating frequencies as follows: Position One - "PRESET", for operation on any one of the predetermined frequencies selected by (1) above; Position Two - "MAN", for operation on any of the manually selected frequencies determined by (4) above; Position Three - "GUARD XMIT", for operation on the guard frequency.

(6) Means for readily setting up new predetermined frequencies on the 20-position switch type control.

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(7) A "SQ DISABLE" switch for disabling the squelch.

3.4.3.7 Vibration - Maximum vibration acceleration to which this control unit shall be subjected shall be governed by curve IV of MIL-E-5400.

3.4.4 Controls, Radio Set C-7307/ARC-51A and C-8616/ARC - These controls shall conform to the applicable requirements of Specification MIL-C-6781 for Type I controls and shall be suitable for installation in a console panel.

3.4.4.1 Function - Radio Set Control C-7307/ARC-51A shall provide prime control of Radio Set AN/ARC-51A or AN/ARC-51AX and shall include all controls for operation and frequency selection of receiver-transmitter RT-743/ARC-51A, RT-780/ARC-51AX, RT-743B/ARC-51A, and RT-780B/ARC-51AX. Radio Set Control C-8616/ARC shall provide prime control of Radio Set AN/ARC-51B or AN/ARC-51BX and shall include all controls for operation and frequency selection of receiver-transmitter RT-767/ARC-51B, RT-742/ARC-51BX, RT-767B/ARC-51B, and RT-742C/ARC-51BX. It shall be possible to manually select any one of the 3500 frequencies and preset up to 20 channels on any of these available frequencies. These controls shall also provide prime control for the associated Frequency Channel Indicator by means of electrical signals, so that the Indicator correctly indicates the frequency or channel selected.

3.4.4.2 Form Factor and Mounting - The overall dimensions of the unit shall not exceed 5-3/4 inches in width, 4-1/8 inches in length, and 6-1/2 inches in depth. As measured from back surface of mounting plate to back of the mating connectors.

3.4.4.3 Weight - The weight of the unit, including the mating connector, shall not exceed 4 pounds, 8 ounces.

3.4.4.4 Contents of the Unit - The control unit shall contain all circuits necessary for (a) to control manual frequency or preset channel selection, emission, ADF operation, and primary power to the Radio Set; and (b) to operate the associated frequency channel indicator.

3.4.4.5 Electrical Connections - All electrical connections to this unit shall be made through connectors conforming to Specification MIL-C-26482.

3.4.4.5.1 Terminations - Radio Set Control C-7307/ARC-51A shall match the 600 ohm audio output impedance of receiver-transmitters RT-743/ARC-51A, RT-743B/ARC-51A, RT-780/ARC-51AX, and RT-780B/ARC-51AX. Radio Set Control C-8616/ARC shall match the 150 ohm audio output impedance of receiver-transmitters RT-767/ARC-51B, RT-742/ARC-51BX, RT-767B/ARC-51B, and RT-742C/ARC-51AX.

3.4.4.6 Controls, Panel-Mounted - The following controls shall be located and suitably marked on the front panel:

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(1) A 20-position rotary switch-type control "PRESET-CHAN" for selecting any one of the 20 predetermined main frequencies.

(2) A 4-position rotary switch for controlling all primary input power to the radio set and for selecting the mode of transmission and reception as follows:

Position One - "OFF";

Position Two - "T/R" operation on frequency selected.

Position Three - "T/R+G" guard receiver operation in addition to that of Position Two;

Position Four - "ADF" operation on frequency selected.

(3) A rotary volume control "VOL" for adjusting the amplitude of the aduio signals delivered to the headset circuit.

(4) Three controls operating a digital display "MC" for manually selecting any one of 3500 channels. The frequency readout display shall indicate the frequency on which the equipment is operating when in the "MAN" position.

(5) A 3-position switch to allow rapid selection of three operating frequencies as follows:

Position One - "PRESET CHAN" for operation on any one of the predetermined frequencies selected by (1) above;

Position Two - "MAN' for operation on any of the manually selected frequencies determined by (4) above;

Position Three - "GUARD XMIT", for operation on the guard frequency.

(6) Means for readily setting up new predetermined frequencies on the 20-position switch type control. Removal of the equipment from the rack or console shall not be required to change the preset frequencies.

(7) A "SQ DISABLE" switch for disabling the squelch.

(8) An "ID-DIM" potentiometer type control for varying the intensity of illumination on the associated

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remote Frequency Channel Indicator.

3.4.4.7 Lighting - Radio Set Control C-7307/ARC-51A shall have either 5 volt or 28 volt red lighting in accordance with MIL-P-7788. Radio Set Control C-8616/ARC-51A shall have 5 volt or 28 volt white lighting in accordance with MIL-P-7788. Lamps shall be replaced from the front panel. Install 28 VDC lamps for shipment and provide 5 VDC Lamps as spares.

3.4.5 Mounting MT-2653/ARC- The mount shall meet the following requirements:

3.4.5.1 Function - The mount shall provide a means of securing the receiver-transmitter to a plane surface within the aircraft or vehicle.

3.4.5.2 Form Factor and Mounting - The mount shall be a sheet metal tinplate with hold-down hardware. Latches shall be provided for engaging with the wing-nut fasteners on the receiver-transmitter. The mount shall be attached to the plane mounting surface by means of eight machine screws. The over-all dimensions shall not exceed 9-1/2 inches wide, 1 inch high by 14-1/4 inches long.

3.4.5.3 Weight - The weight shall not exceed 0.5 pound.

3.4.5.4 Safety Wire Hole - The safety wire hole in the mounting shall be so designed as to eliminate possibility of curring wire during use.

3.4.6 Cooler, Air, Electronic Equipment HD-513/ARC and HD-615/ARC-51X cooler shall meet the following requirements:

3.4.6.1 Function - The cooler shall provide a circulation means for external air when the radio set is operating in a self-cooling mode.

3.4.6.2 Form Factor and Mounting - The cooler shall be a single subassembly capable of being attached to the rear of the receiver-transmitter by means of captive screws. It shall be possible to remove the cooler from the receiver-transmitter using ordinary hand tools and without requiring removal of any cables. The over-all dimensions of the unit shall not exceed 6-5/32 inches wide, 6-3/8 inches high and 3-1/8 inches long.

3.4.6.3 Weight - The weight of this unit shall not exceed 1.1 pounds.

3.4.6.4 Contents - This cooler shall contain a blower and air transition means for circulating ambient air through the external cover of the heat exchanger on the receiver-transmitter. The cooler shall be powered and controlled by means of a connector mating with the rear connector of the receiver-transmitter. HD-513/ARC and HD-615/ARC-51X shall operate with their respective receiver-transmitters as listed in paragraph 1.3.

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3.4.6.5 Electrical Connections - All electrical connections shall be made through a connector conforming to Specification MIL-C-26482.

3.4.6.6 Vibration - This cooler shall be subjected to vibration only when attached to the receiver-transmitter.

3.4.7 Indicator, Standing Wave Ratio ID-1003/ARC- The indicator shall meet the following requirements:

3.4.7.1 Function - The indicator shall be used to measure incident and reflected power of the receiver-transmitter and provide a low-pass filter.

3.4.7.2 Form Factor and Mounting - The reflectometer filter unit shall be contained in a rectangular box capable of being attached to the front of the receiver-transmitter by means of captive screws. It shall be possible to remove the unit from the receiver-transmitter using ordinary hand tools and without requiring removal of any cables except the pendant cable attached to the antenna jack of the receiver-transmitter. The overall dimensions of the unit shall not exceed 4-3/4 inches wide, 1-7/16 inches high by 3-3/4 inches long.

3.4.7.3 Weight - The weight of this unit shall not exceed 1.1 pounds.

3.4.7.4 Contents - The indicator shall contain a meter, switch and all circuits necessary to measure the incident and reflected power of the receiver-transmitter and attenuate all emitted antenna signals above the highest carrier frequency.

3.4.7.5 Electrical Connections - All electrical connections to the unit shall be made through connectors conforming to Specification MIL-C-26482.

3.4.7.6 Controls, Panel-Mounted - The following control shall be located and suitably marked on the front panel:

(1) (1) A momentary push-button switch to be depressed for measuring reflected power by means of the meter.

3.4.7.7 Vibration - The indicator shall be subjected to vibration only when attached to the receiver-transmitter.

3.4.8 Indicators, Frequency Channel ID-1472/ARC-51A and ID-1752/ARC - The indicators shall meet the following requirements:

3.4.8.1 Function - These indicators, when mounted in a remote position in an aircraft, shall follow the switching action of Radio Set Controls C-7307/ARC-51A and C-8616/ARC by indicating the frequency to which the Radio receiver-transmitter is tuned when manual tuning is used, or the channel when preset channel operation is used.

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3.4.8.2 Form Factor - The Indicators shall be enclosed in a round case meeting the requirements of Military Standard MS33639(ASG) for nominal 2-inch case size. The Indicator shall be compatible for use with a front Aircraft Instrument Mounting flange in conformance with Military Standard MS28055-6. The length of the Indicator shall not exceed 3-3/16 inches from the back mounting surface of the mounting flange to the end of the connector receptacle (excluding mating connector).

3.4.8.3 Total Weight - The total weight of the equipment excluding cables shall not exceed 1.5 pounds.

3.4.8.4 Contents - The Indicator shall contain a sector coil, digit wheel, and all circuitry necessary to provide a remote frequency channel readout which follows the switching action of Radio Set Controls C-7307/ARC-51A and C-8616/ARC.

3.4.8.5 Operation - The Indicator shall, when mounted in a position remote to an aircraft transceiver, provide the following information with respect to the transceiver control box settings:

Transceiver Mode	Remote Indication
Power ON	Sufficient for instantaneous and unmistakeable recognition.
Manual Frequency Selection	Selected frequency will be shown in megacycles by a five digit presentation (See 3.4.1).
Pre-set Channel	Pre-set channel will be shown by one or two digits. Unused spaces will be blank.
Guard Transmit	Letter G will be shown. Unused spaces will be blank.

3.4.8.6 Remote Presentation - In the transceiver frequency range of 225.00 to 399.95 megacycles, the Indicator shall show frequency in megacycles in .05 megacycle increments corresponding to the arithmetic series 225.00, 225.05, 225.10 ——— 399.95. Additionally, it shall be capable of indicating at least 20 channels during pre-set channel operation and one guard channel during guard transmit operation.

3.4.8.7 Independent Operation of Transceiver - Normal transceiver operation shall not be affected when the remote indicator is disconnected, inoperative or fails.

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3.4.8.8 Indicator Operation and Changeover Time - The remote presentation shall instantaneously be activated and follow the switching action (clockwise or counterclockwise) of the associated transceiver power, preset channel, and frequency control switches. All numerals and characters shall reach their proper register within 2 seconds after switching under standard conditions.

3.4.8.9 Damping - The movements of the display wheels shall be damped to a basic value of not less than 20% critical damping. At room temperature and power input of 27.5 volts, the damping shall be such that after a change over six positions, the first overshoot beyond the desired numeral shall not exceed 1.5 heights of the display digit.

3.4.8.10 Lighting - Indicator ID-1472/ARC-51A shall have either 28 VDC or 5 VDC red lighting in accordance with MIL-P-7788. Indicator ID-1752 /ARC shall have either 28 VDC or 5 VDC white lighting in accordance with MIL-P-7788. Lamps shall be accessible and easily replaceable from the front of the unit. A separate board may be used containing lamps and associated wiring. The lighting shall be adjusted in intensity by the dimmer control on Radio Set Controls C-7307/ARC-51A and C-8616/ARC.

3.4.8.11 Cooling - The design for cooling the equipment shall comply with the requirements of Specification MIL-E-5400. The thermal design tests shall be conducted as required by Specification MIL-T-23103.

3.4.8.12 Vibration - The equipment shall meet the vibration requirements of Curve IV of Specification MIL-E-5400.

3.4.8.13 Degraded Performance - Degraded performance will be permitted for all voltage transients from 0 to 0.5 sec duration as shown on figure 2 of MIL-STD-704 for Normal Electric System Operation except performance shall not be affected by short pulses of less than 50 micro-seconds. Operation shall return to normal when the 0 to .5 sec transient expires.

3.4.8.14 Electrical Connections - All electrical connections to this unit shall be made through a connector conforming to Specification MIL-C-26482.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction (First Article) Tests

(2) Initial Production Tests

(3) Acceptance Tests

(4) Life Tests

4.2 Preproduction (First Article) Tests - Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under the approved test procedure of 4.6. The Government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.2.1 Preproduction (First Article) Test Data - The contractor shall submit all data collected in conducting these tests to the procuring activity for review and approval. On first production contracts this data shall include a list of all electrical and electronic parts giving their specified voltage, current and temperature rating and the applied circuit voltage, current and ambient and surface temperatures. The ambient and surface temperatures shall be obtained under the extreme high temperature operating condition.

4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with the procedures of Specification MIL-T-5422 and interference tests in accordance with Specification MIL-I-6181.

4.2.3 Preproduction (First Article) Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract.

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4.2.4 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection and for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Initial Production Tests - One of the first ten production equipments shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests. The preproduction sample shall not be selected for this test.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance and quality and that it meets all applicable requirements.

4.3.2 Accessory Material - In addition to the complete equipment submitted for Initial Production Tests the contractor shall also submit such accessory material and data necessary to test the equipment.

4.3.3 Initial Production Sample Approval - Approval of the Initial Production Sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the Initial Production Sample to pass any of the tests shall be cause for deliveries of the equipment under the contract to cease until proper correction action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.4 Reconditioning of Initial Production Test Sample - On completion of the initial production test the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government Inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showning quantative results for all acceptance tests. Such reports shall be signed by an authorized representative of the

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contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Reliability Assurance Tests

(4) Special Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Run in Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Run in Test - Each equipment shall be operated under the conditions specified herein for a period of ten hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment. This test shall be deleted if the reliability test includes a test on each equipment which consumes at least ten hours of operation

Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ±3 g's

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The equipment shall be vibrated (without vibration isolators) for a period of ten minutes prior to the beginning of the ten hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for five minutes and lateral to the plane for five minutes. Where it is not feasible to vibrate the equipment in two directions the vertical direction shall be used. During the ten hour period of operation following the ten minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the ten minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The ten hour period specified above may be composed of two five hour periods to conform with standard working hours.

4.4.2 Sampling Tests - Equipments selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the Government inspector in accordance with the following: (Sampling tests shall be conducted only when Reliability Assurance Tests of 4.4.3 are deleted by contractual action.)

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Sampling Test
First 10	0
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

NOTE: If by contract action the Initial Production Test (4.3) is deleted, a Sample Test shall be conducted on one equipment from the first 10 produced.

4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction initial production, individual and special tests.

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(3) Manufacturing run in test specified in 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

4.4.3 Reliability Assurance Tests - Reliability Assurance Tests shall be conducted using MIL-STD-781. Tests as required by both the Qualification Phase and the Sampling Phase shall be conducted.

4.4.3.1 Qualification Phase - Prior to the acceptance of equipments under the contract or order, a minimum of three (3) equipments shall be tested as outlined in MIL-STD-781, under the section entitled "Qualification Phase of Production Reliability Tests." The maximum number of equipments to be used shall be those listed in Table 5 of MIL-STD-781. For the Qualification Phase, Test Level E shall be used. Test Plan II of MIL-STD-781 and an MTBF of 300 hours shall be used to determine the accept-reject criteria. When the contractor has been in continuous production of AN/ARC-51 equipments during the preceeding six months, the "Qualification Phase" of the reliability assurance tests may be omitted.

4.4.3.2 Reliability Sampling Phase Tests - Samples of the production equipment shall be tested as outlined in MIL-STD-781 (as modified herein), under the Section entitled "Sampling Phase of Production Reliability Tests." For the Sampling Phase Test Level E shall be used.

4.4.3.2.1 All Equipment Test - Each equipment produced except those submitted for the Reliability Qualification Test, shall be tested for 24 hours. Prior to the 24 hour test on each equipment, a burn-in period may be used at the option of the contractor. If the burn-in period is to be used the details thereof must be included in the approved test procedures. To determine whether the MTBF is being met at any time during the contract the operating test hours and the failures thereon (not counting burn-in failures or burn-in operating time) shall be totaled and the results compared with the reject line of Test Plan II of MIL-STD-781. (Extend the line as necessary to accommodate the data.). These totals shall accumulate so that at any one time the experience from the beginning of the contract is included. At the conclusion of each month the test results shall be sent to the procuring activity and to the Naval Air Systems Command, Attention: Avionics Division. At any time that the current totals of test hours and test failures plotted on Test Plan II curves show a reject situation, the procuring activity shall be notified. The procuring activity reserves the right to stop the acceptance of equipment at any time that a reject situation exists pending a review of the contractor's efforts to improve the equipment, the equipment parts, the equipment workmanship, etc., so that the entire compilation will show other than a reject decision.

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4.4.3.3 Test Details - The test details such as the length of the test cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria, preventive maintenance to be allowed during the test, etc., shall be part of the test procedures to be submitted and approved by the procuring activity prior to the beginning of the Qualification Test Phase of the Reliability Assurance Tests.

4.4.4 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests.

4.4.4.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early equipment after an engineering or material change.

(2) Whenever failure reports or other information indicate that additional tests are required. (This will be determined by the procuring activity.)

4.4.4.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.4.5 Equipment Failure - Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

(1) Determine the cause of failure

(2) Determine if the failure is an isolated case or design defect

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

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4.5 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing the life tests. The test shall be of 300 hours duration and shall be conducted on equipments that have passed the individual test. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the Government Inspector in accordance with the following: (Equipments which have successfully passed the Initial Production Test, Sampling Tests, Reliability Tests, or Special Tests may be selected for life tests.) When reliability tests are conducted, the life test shall be omitted if, during the reliability tests, a quantity of equipments equal to, or more than, that listed below receive at least 300 hours each of test time.

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Life Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions:

Temperature	Normal room
Altitude	Normal ground (0-5000 ft)
Humidity	Room ambient
A.C. Voltage	115 ±5 volts (at lowest applicable frequency)
D.C. Voltage	27.5 ±2.0 volts

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check - At approximately 8 hours intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

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(1) Part type number

(2) The circuit reference symbol number

(3) The part function

(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure

4.5.4.1 Failure Report - In the event of a failure, the Government Inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the Government Inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts, such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.7 Reconditioning of Tested Equipment - Equipment which has been subjected to initial production acceptance and life tests shall be reconditioned by the contractor by replacing all wear or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.8 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.9 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Government Inspector.

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5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-E-17555 and MIL-STD-794. In the event the equipment is not covered in Specification MIL-E-17555, the method of preservation for Level A shall be determined in accordance with the selection chart in Appendix D of MIL-STD-794.

6. NOTES

6.1 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 27.5 +0.5 V DC and 115 +1.0 V AC. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standard of performance.

6.2 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Reference Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.3 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shws advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

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6.4 Ordering Data - Procurement documents should specify the following:

(1) Title, number and date of this specification

(2) List of items (1.2).

(3) Selection of applicable levels of preservation, packaging and packing (see 5.1).

6.5 As a general rule non-repairable subassemblies should be encapsulated or hermetically-sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the sub-assembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies), the non-repairable subassembly should evidence a Mean-time-to-failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

Project No. 5821-N076

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Naval Air Systems Command
Washington, D.C. 20360

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-R-22659D(AS) RADIO SETS AN/ARC-51A, AN/ARC-51B, AN/ARC-51AX, and AN/ARC-51BX

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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